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D20A - 44

BlockChain Assignment 1

One-Word Answerable Questions

Bitcoin Basics

Bitcoin was introduced by whom? Satoshi Nakamoto

Bitcoin transactions are recorded on which data structure? Blockchain

Bitcoin uses which consensus mechanism? Proof of Work

The smallest unit of Bitcoin is called? Satoshi

Bitcoin is a peer-to-peer type of what system? Payment system

Wallets

A wallet stores the user's Private key.

The recovery phrase used in wallets is called a Seed phrase.

Hardware wallets store keys Offline (online/offline).

Bitcoin addresses are derived from Public keys.

What is required to sign a transaction? (public/private) Private

Security

Bitcoin prevents double-spending using Blockchain.

Bitcoin is secured using Asymmetric cryptography (asymmetric/symmetric).

To run a Bitcoin node, the software required is Bitcoin Core

The process of verifying transactions is called Validation

The Bitcoin network is described as Decentralized (centralized/decentralized).

Mining

Miners solve Cryptographic puzzles to add blocks.

Mining rewards a block with newly created Bitcoins

The block time target for Bitcoin is 10 minutes.

Mining difficulty adjusts every 2016 blocks.

Miners compete to find a valid block Hash (hash/signature).

Blockchain Structure

Each block contains a Merkle root.

Blocks are linked using the previous block's Hash

A Bitcoin transaction contains inputs and Outputs

Unspent transaction outputs are called UTXOs

The first-ever Bitcoin block is called the Genesis block.

Network

A full node stores the entire blockchain History

Bitcoin uses which protocol for communication? (TCP/P2P) TCP

A light wallet uses SPV mode (SPV/full).

Nodes relay information across a Peer-to-peer network.

The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Block height: 961,122

Block hash: 00000...8a6be

Timestamp: 05 August 2026, 11:38:03 AM

Number of transactions: 4,629

Miner/pool name: Unknown

Task B2: Explore a Transaction

Write the TXID of the transaction. 7cf9...36c7

Number of inputs: 1

Number of outputs: 8

Transaction fee: 0 BTC (0 satoshis)

How many confirmations does it have? 2

Task B3: Explore an Address

Record (Bitcoin Address): 1K6KoYC69NnafWJ7YgtrpwJxBLijWqwa6

Total BTC received: 0.12772136 BTC

Total BTC sent: 0.02073492 BTC

Current balance: 0.10698644 BTC

What is the role of a full node? Validates transactions and blocks independently, stores the complete blockchain, and relays data.

List any two developer resources available on bitcoin.org. Developer Guide; Developer Reference

Explore:

Task C1: Market Information

Current price of BTC: \$64,197.87

24-hour % change: +0.94%

Market capitalization: \$1.28T

Circulating supply: 20.06M BTC

Total/maximum supply: 20.06M BTC / 21M BTC

Task C2: Historical Data

What was the price of Bitcoin one year ago? \$115,000

Highest price in the last 30 days: \$66,800

Lowest price in the last 30 days: \$62,000

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile? Its price changes rapidly due to demand, investor sentiment, news, regulation, and market conditions.

List two investment risks mentioned. Price volatility; Regulatory uncertainty

Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

- bitcoin.org because it provides clear, official explanations of Bitcoin concepts.

In one paragraph, explain how blockchain ensures trust without a central authority.

- Blockchain uses a distributed ledger, cryptographic security, and consensus to verify transactions. Every block is linked to the previous one, making records difficult to alter. Since many independent nodes validate the blockchain, no central authority is needed to establish trust.